

Nafiul Islam Naim

B.Sc. (Honours), 4th-Year Student
Department of Applied Mathematics
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🎓 Education

Bachelor of Science Oct 2022 – Expected Dec 2026
Department of Applied Mathematics, University of Dhaka, Dhaka, Bangladesh
CGPA: 3.00/4.00 (through 3rd year)

Undergraduate Project : “ Principal Component Analysis and Its Applications ” — Developed a PCA-Based machine learning framework for predicting annual Heat Wave Days (HWDs) across Bangladesh using ERA5 reanalysis climate data.

Supervisor: [Shuchi Chaki](#)

🔬 Research Interests

My research interests focus on **Deep Learning and Foundation Models, Computer Vision and Multimodal AI, AI Agents and Autonomous Systems, Robotics and Physical AI, AI for Climate and Earth System Science, Data Science and Statistical Learning, and Optimization and Scientific Computing**. I am looking to work with complex data and applying computational and AI techniques to find solutions to real-world problems, specifically finding efficient, resource-limited, and interpretable solutions.

📖 Research Project (Ongoing)

BD-HeatPred: A Multi-Source Daily Predictor Dataset for Heatwave Research in Bangladesh, February–June, 1991–2025

🔗 [project page](#)

- Developing a region-specific dataset for Bangladesh by integrating 35 atmospheric, land-surface, and global teleconnection predictors across five regions: Dhaka, Khulna, Rajshahi, Rangpur, and the Rangamati–Bandarban–Khagrachari (RBK).

BD-HeatPred-Based Heatwave Onset Prediction and Regional Driver Analysis in Bangladesh

- Heatwave onset prediction in Bangladesh using the BD-HeatPred dataset, with regional driver analysis to identify the most influential predictors across five regions :Dhaka, Khulna, Rajshahi, Rangpur, and the Rangamati–Bandarban–Khagrachari (RBK).

</> Projects

Principal Component Analysis and Its Applications

🔗 [GitHub](#)

- Developed an interpretable PCA-based machine learning framework to predict annual Heat Wave Days (HWDs) across Bangladesh's eight administrative divisions using ERA5 atmospheric data .
- Evaluated the effectiveness of PCA for dimensionality reduction and prediction performance, and applied SHAP-based explainable AI to identify dominant atmospheric drivers associated with heat-wave occurrence across Bangladesh.

Tools: Python, Pandas, NumPy, Scikit-learn, PCA, mRMR, SHAP, ERA5

Technical skills

Programming	Python, C, C++, Dart , MATLAB
Data Analysis & Visualization	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
ML & Deep Learning	PyTorch, Keras, Explainable AI (SHAP), PCA, mRMR
Computer Vision	Ultralytics YOLO, OpenCV, Roboflow
Climate & Geospatial Analysis	ERA5/ERA5-Land, NetCDF, Xarray, GeoPandas, GIS, Spatial Analysis, Grid-Based Analysis, Remote Sensing (NDVI, NDWI), Google Earth Engine(GEE), Climate Anomaly & Teleconnection Analysis(SST, ENSO, IOD, MJO, OLR)
Web & Application Development	HTML, CSS, Flutter
Tools	Git, GitHub, VS Code, PyCharm, LaTeX, Google Colab, Jupyter Notebook, Canva

Standardized Tests

I plan to take the **International English Language Testing System (IELTS)** and the **Graduate Record Examination (GRE)** soon.

Referees

- **Shuchi Chaki**

Graduate Teaching Assistant

North Carolina State University

Email: shuchi@du.ac.bd

Lecturer (on study leave)

Department of Applied Mathematics, University of Dhaka, Dhaka, Bangladesh

- **Kawser Ahammed**

PhD Student (fifth year)

Department of Electrical and Computer Engineering

Old Dominion University, Virginia, USA

Email: kaham001@odu.edu

[Website](#)

Assistant Professor (on study leave)

Department of Electrical and Electronic Engineering (EEE),

[Jatiya Kabi Kazi Nazrul Islam University, Trishal, Mymensingh, Bangladesh](#)